

Study Report: Advanced Apex Programming - Salesforce CRM Developer / Administrator

Written by Gagan Pasupuleti

Family: Salesforce CRM Developer / Administrator
Level: Advanced
Chapters: 7
Source: Reference: Advanced Apex Programming - Dan Appleman

Official sources and free libraries

- <https://www.salesforce.com/blog/advanced-apex-programming/>

Summary

Study report for **Advanced Apex Programming** by Dan Appleman (Advanced level) mapped to the Salesforce CRM Developer / Administrator role. Bulk-safe Apex, triggers, and governor limit patterns.

Chapters

Track: Book-Intro

Chapter 1: About Advanced Apex Programming [Advanced]

Chapter focus: About Advanced Apex Programming** *(Level: Advanced)

This study report summarizes **Advanced Apex Programming** by Dan Appleman for the Salesforce CRM Developer / Administrator role. The resource is rated Advanced level. Bulk-safe Apex, triggers, and governor limit patterns. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Advanced Apex Programming
# Author: Dan Appleman
# Role: Salesforce CRM Developer / Administrator
# Level: Advanced
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Advanced Apex Programming.

When to use it

When learning Salesforce CRM Developer / Administrator skills at Advanced level.

Where to use it

Salesforce CRM Developer / Administrator training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Advanced]

Chapter focus: Why This Book Matters for Your Role** *(Level: Advanced)

Salesforce CRM Developer / Administrator professionals use ideas from Advanced Apex Programming to solve real workplace problems. Bulk-safe Apex, triggers, and governor limit patterns. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Salesforce CRM Developer / Administrator

Book focus: Bulk-safe Apex, triggers, and governor limit patterns.

Recommended level: Advanced

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Salesforce CRM Developer / Administrator bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a advanced skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Advanced]

Chapter focus: Key Topics Covered** *(Level: Advanced)

The main topics in Advanced Apex Programming include practical concepts described as: Bulk-safe Apex, triggers, and governor limit patterns. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Salesforce CRM Developer / Administrator jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Asynchronous Apex Patterns [Advanced]

Chapter focus: Asynchronous Apex Patterns** *(Level: Advanced)

Queueable, Batch Apex, and Schedulable process large volumes beyond synchronous limits. Batch implements Database.Batchable with scope chunks. Platform Events decouple publishers and subscribers for event-driven scale.

Code Reference:

```
public class EnrollmentBatch implements Database.Batchable<SObject> {
    public Database.QueryLocator start(Database.BatchableContext bc) {
        return Database.getQueryLocator('SELECT Id FROM Enrollment__c');
    }
}
```

What it shows: Batch skeleton processes custom enrollment objects nightly.

Topic guide

What it actually is

Async Apex handles enterprise data volume outside user transactions.

When to use it

Nightly syncs, rollups, and data archival.

Where to use it

Orgs exceeding synchronous governor thresholds.

Real use example

Batch recalculates district utilization scores for 2M child records.

Chapter 5: Applied: Enterprise Integration Architecture [Advanced]

Chapter focus: Enterprise Integration Architecture** *(Level: Advanced)

Apply Salesforce Well-Architected: reliable, secure, adaptable. Use anti-corruption layers, outbox patterns, and idempotent consumers. MuleSoft or Event Relay for hybrid buses.

Code Reference:

```
// Change Data Capture subscriber processes Account updates to ERP
```

What it shows: CDC comment illustrates event-driven integration style.

Topic guide

What it actually is

Integration architecture prevents brittle point-to-point spaghetti.

When to use it

Multi-system landscapes with ERP, LMS, and billing.

Where to use it

Large enterprises and acquisition-heavy portfolios.

Real use example

CDC streams opportunity wins to provisioning microservice within seconds.

Chapter 6: Applied: Advanced Security and Shield [Advanced]

Chapter focus: Advanced Security and Shield** *(Level: Advanced)

Shield Platform Encryption, Event Monitoring, and Field Audit Trail address compliance. Transaction Security policies block CSV exports by profile. Monitor Login Forensics for credential stuffing. Tenant secret rotation requires planning.

Code Reference:

```
// Event Monitoring: ReportEvent for bulk data export detection
```

What it shows: ReportEvent tracks who exported large contact lists.

Topic guide

What it actually is

Advanced security features protect regulated CRM data.

When to use it

Healthcare, financial services, and public sector CRM.

Where to use it

Orgs under HIPAA, PCI, or FedRAMP requirements.

Real use example

Transaction Security blocks guest user SOQL injection attempt.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Advanced]

Chapter focus: Study Plan and Practice** *(Level: Advanced)

Finish Advanced Apex Programming with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Salesforce CRM Developer / Administrator. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a advanced project screenshot.